Project Design Phase-II / Requirement Analysis

Technology Stack (Architecture & Stack)

Date	28 October 2026
Team ID	Ujjawal Rathod (Individual Capstone)
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	2 Marks

Define Your Technology Stack: Identifies the languages, data storage, ingestion connectors, analytical transformations, BI visualization tools, and cloud web framework used to deliver the Electric Vehicle Market Performance & Insights solution.

S.No	Component	Description	Technology	Justification / Purpose
1.	**Programming / Query Language**	Languages and expressions used to query, calculate metrics, and build the web application interface.	**Tableau Calculated Fields (FIXED LOD), Python 3.14, HTML5, CSS3**	Used Tableau LOD formulas for accurate market share across 200 manufacturers, and Python (Flask) + HTML5/CSS3 to build the responsive web portal.
2.	**Database / Data Warehouse**	Structured data storage holding the 300,000 electric vehicle records across 37 attributes.	**CSV Flat Files (`ev_market_train.csv` & `ev_market_test.csv`) + Tableau Data Extract (`.twbx`)**	Stores 200,000 training rows and 100,000 test rows locally and packages them into a compressed, high-speed Tableau Packaged Workbook (`.twbx`).
3.	**Data Ingestion**	Mechanism used to connect, combine, and load the split datasets into the BI environment.	**Tableau Native Text File Connector & Union (Specific/ Manual Union)**	Combines `ev_market_train.csv` and `ev_market_test.csv` from a single directory into a unified 300,000-row table with automatic Table Name tracking.
4.	**Data Transformation and Business Logic**	Tools and rules used to clean data types, assign geographic roles, filter Top-N segments, and compute KPIs.	**Tableau Public Data Source Prep, Quick Table Calculations & FIXED LOD Expressions**	Sets Model Year to Discrete, maps Manufacturing Country to Country/Region, filters Top 10 Manufacturers / Top 5 Categories, and computes Market Share %.
5.	**Data Visualization / Analytics**	BI platform used to construct the 9 interactive visualizations, persona dashboards, and data story.	**Tableau Public (Desktop & Cloud)**	Enables rapid creation of V1–V9 charts, a 3×3 master dashboard with cross-filtering, 3 persona mini-dashboards, and a 4-point executive Story.
6.	**Front-End / User Interface**	Web interface and hosting stack through which	**Python Flask, Jinja2 (`index.html`), GitHub & Vercel Cloud**	Flask renders the web UI containing live Tableau Public iframes, version-controlled on

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		stakeholders access the embedded dashboard and story.		GitHub and deployed serverless on Vercel for instant browser access.